From Footage to Ethogram: A Deployable Pipeline for Continuous Behavioural Monitoring of a Captive Octopus

Siddharth Raj*
University of Chicago
Chicago, IL, USA
siddharthraj000@gmail.com

Krishan Mohan Patel*
Institute for Digital Media Education
University of Teacher Education Styria
Graz, Austria

Himanshu Kumar
Research and Development
Arishna IoT Solutions
Bangalore, India
ORCID: 0009-0005-2499-2715

Rishabh Thakur
Research and Development
Arishna IoT Solutions
Bangalore, India
rishabh.thakur.hp19@gmail.com

Shivansh Pachnanda
University of Delhi
New Delhi, India

Harendra Pal Singh
Cluster Innovation Center
University of Delhi
New Delhi, India

Harald Burgsteiner
Institute for Digital Media Education
University of Teacher Education Styria
Graz, Austria
ORCID: 0000-0001-7800-8414

Wolfgang Slany
Institute of Software Engineering and AI
Graz University of Technology
Graz, Austria
ORCID: 0000-0002-4979-6156

***Abstract*—Continuous behavioural monitoring is the obvious use for an aquarium's cameras and remains out of reach: manual ethogram scoring is bounded by expert hours, and prompting a large vision–language model (VLM) on every clip fails on arithmetic rather than on accuracy. The 892 h we hold for a single adult *Octopus vulgaris* is about 160,000 twenty-second clips, so one five-pass labelling run costs 372 to 1,030 h of wall-clock on an API the aquarium does not have. We instead treat a 235B-parameter VLM as an *offline teacher* and distil it into a cascade of small models that runs on site. Cheap gates reject 60% of videos without a full decode, leaving only about 60 h ever decoded; 42.7% of surviving clips still contain no animal, which is why the expensive model is spent last rather than first. Every gate is placed by a measurement on human-labelled data rather than by intuition, and in each case the measurement changed the design. The students reach 0.665 macro-F1 on a six-class ethogram against a majority baseline of 0.100, and 0.642 mask IoU from 3.2 M parameters with no prompt at inference, above the 0.374 its teacher architecture reaches zero-shot per frame. Applied to the corpus, the pipeline recovers a circadian profile peaking at 45% visible activity at 17:00 against 1 to 5% overnight, and shows human presence nearly doubling movement. Arm-tip speed, computed from the masks without ever seeing the teacher's answer, independently recovers the same ordering of behaviours. We release 5,222 teacher-labelled clips with full vote distributions, about 25,000 captions, 969 human labels and masks, and three frozen video-level benchmarks.**

***Index Terms*—cephalopods, ethogram, knowledge distillation, vision–language models, video segmentation**

*Siddharth Raj and Krishan Mohan Patel contributed equally (shared first authorship).

I. Introduction

An aquarium camera produces footage continuously; turning it into behavioural knowledge does not scale the same way. Manual ethogram scoring is the accepted standard and is bounded by expert hours. The obvious modern substitute—prompt a large VLM on every clip—fails on arithmetic rather than on accuracy. The 892 h of colour-camera footage we hold for a single animal is ≈160,000 twenty-second clips; at the throughput we measured against a hosted 235B model (13–36 clips/min sustained), a five-pass labelling run is 372–1,030 h of wall-clock, and it requires an API the aquarium does not have. That figure is for *one* animal: the same archive holds recordings of four others, about three times as much footage again, so the cost scales with every animal a facility monitors.

The waste is also structural, not merely large. Cephalopods spend most of their time hidden or motionless: after the extraction gates described below, 42.7% of surviving clips still contain no visible animal and a further 16.7% show it stationary. Prompting a frontier model uniformly spends most of its budget confirming that nothing is happening.

We therefore treat the VLM as an *offline teacher* and build a cascade of small models that runs locally. The contribution is not the cascade as an architecture—teacher–student distillation is standard—but that every gate in it is placed by a measurement on human-labelled data, and that several of those measurements contradicted the intuitive choice. We then show the pipeline produces behavioural findings, which is the point of building it, and release the labels, captions and benchmarks as a resource for future work on octopus behaviour—a field with no public annotated corpus to build on.

II. Related Work

Our own prior work established the recording setup and the framing of inter-species communication with octopodes [1]. For perception we use CLIP [2] features, Grounding DINO [3] and SAM2 [4] as annotators, LoRA [5] for adaptation, MobileNetV3 [6] as a deployable backbone, and Qwen3-VL [7] as the teacher.

Foundation models as teachers, not detectors. Open-vocabulary detection is attractive here because no octopus detector exists off the shelf. In practice OWLv2 [8] reported an octopus in 231 of 232 frames at its default threshold. Its scores are not uninformative—they rank the two classes at AUC 0.759 against human adjudication—but the distributions overlap so heavily that no threshold is usable: at its own optimum it still passes 30% of the frames we wanted removed while discarding 32% of the real animals. A weak ranking signal with no viable operating point is not a detector, which is the case for distillation rather than prompting.

Automated behaviour analysis, and why pose-first does not transfer. Quantitative ethology has largely been automated by markerless pose estimation: DeepLabCut [9] and SLEAP [10] fit user-defined keypoint skeletons and are the default tools for rodents, insects and fish. That formulation assumes a body plan with a fixed, identifiable set of landmarks. An octopus has no rigid skeleton, eight arms that are mutually indistinguishable, and a deformable mantle, so "keypoint k" is not well defined across frames and the arms cannot be consistently ordered. We therefore work from silhouettes rather than landmarks, and treat the anatomical graph (Sec. VII) as derived from the mask rather than fitted directly. The behavioural vocabulary itself is not ours to invent: body-pattern and action catalogues for cephalopods [11] and a published ethogram for benthic octopods [12] define the categories, and our six-class sheet is a coarsening of them chosen for what is separable from the fixed tank cameras at 20 s granularity. Distillation follows the standard recipe [13], with the teacher a prompted VLM rather than a trained classifier.

Octopus segmentation. The closest work is HideAndSeg [14], a concurrent annotation-light pipeline: a human annotates initial frames to prompt SAM2, the resulting masks train a YOLOv11 detector, and that detector auto-prompts SAM2 over the remaining footage. Its central insight—re-detecting per frame so a contaminated propagation track can recover—is correct, and our seed-confidence gate serves the same purpose. Its field coverage substantially exceeds ours: 148 videos and 366,514 frames of wild juvenile *O. insularis* across three Brazilian sites, against our single individual in one tank. We compare against it in Table IV, where the condition rows matter more than the metric row. Notably, its stated motivation—the absence of publicly available annotated oc-

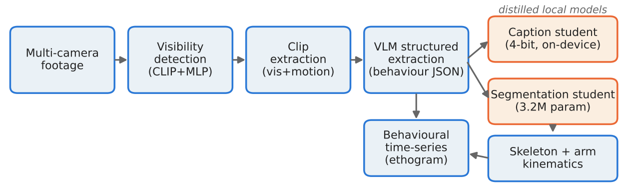

Fig. 1. The pipeline. Cheap local gates run first; the teacher VLM is reached only by clips that survive them, and is used offline to train the students that are actually deployed.

TABLE I
The Cascade: Survival Rates and Gate Placement

Funnel (1,769 videos, 892.5 h)	
probe	1,054 videos (60%) discarded on ≈10 seek frames, with no full decode
decoded at all	**≈60** h
clips to VLM	5,222
no animal	42.7%
stationary	16.7%
active	**40.6%**
Gate placement	
presence	the VLM returns *not present* on 63% of clips the trained detector had marked verified. Prevented trusting the detector.
label confidence	human agreement tracks the five-vote margin: 0.73 unanimous, 0.86 at four-of-five, 0.43 at ≤three-of-five. Prevented arg-max labels teaching false confidence; targets are the vote distribution.
motion	per-video *normalised* motion passes static scenes—a flickering lamp normalises up to "motion". Absolute changed-pixel fraction used instead.

topus video—holds after publication, since it releases neither data nor code; our release is a partial answer to it.

III. Pipeline and the Measurements That Set It

The pipeline has two halves (Fig. 1). A *selection* cascade decides what is worth looking at closely, running cheapest-first: a frame-level probe over sparsely seeked frames, an absolute-motion gate, a trained presence probe over the surviving windows, and only then the teacher VLM. An *extraction* stage then runs the distilled students over what survives—a six-class ethogram label, a one-sentence caption, and a pixel mask (Sec. IV)—and the mask in turn supports an anatomical mantle–head–arm graph and per-arm kinematics (Sec. VII). The behavioural time series in Sec. V is assembled from those outputs. Only the selection half has gates, and it is where the measurements below apply. Table I reports how much footage survives each stage and, for every gate, the measurement that placed it alongside the decision that measurement reversed. In each case the measurement changed the design we would have chosen without it.

Labelling with five disjoint views. Each surviving clip is sent to the teacher five times. Pass p offsets its uniform frame

TABLE II
ETHOGRAM CLASSIFICATION RESULTS

model	macro-F1	accuracy
majority class	0.100	43.1%
pooled CLIP $\rightarrow$ linear	0.400	–
CLIP $\rightarrow$ BiGRU	0.530	–
three self-supervised backbones, ensembled	0.617	71.4%
+ **mask-guided crop (five members)**	**0.665**	**75.4%**
per class (precision / recall)		
no octopus	0.93 / 0.90	
resting	0.71 / 0.60	
exploration/manipulation	0.66 / 0.67	
locomotion	0.64 / 0.73	
reaching out of water	0.57 / 0.75	
human/enrichment interaction	0.44 / 0.43	

grid by $(p-1)\,\mathrm{step}/5$, so the five samplings are disjoint and their union tiles every dense frame; a clip therefore receives five independent presence votes, ethogram votes and captions. This is not redundancy for its own sake: the majority vote differs from a single pass on 705 clips (13.5%), 36.2% of clips are split rather than unanimous, and—per Table I—the margin predicts human agreement, which is what licenses training on soft targets. Voting only reduces variance if the passes are exchangeable, so we checked that rather than assuming it. On the 1,144 clips with all five passes, pairwise Cohen κ across all ten pass pairs spans just 0.617–0.653: no pass is an outlier. Agreement itself is moderate on the six-class label (Fleiss $\kappa = 0.634$, mean pairwise 0.729) and strong on presence ($\kappa = 0.818$, 0.910)—that is, the teacher is far more certain *that* the animal is there than *what* it is doing, which is precisely why presence is used as a gate and behaviour is carried as a distribution.

IV. LOCAL MODELS

Three students consume the teacher's output. All are trained with splits *by source video*; a held-out clip from a training video is not held out in any useful sense.

Ethogram classifier. Six classes in one head—*no octopus*, resting, exploration/manipulation, locomotion, reaching out of water, human/enrichment interaction, where locomotion merges the teacher's *crawling* and *swimming/jetting* that Fig. 3 and Table V report separately—because that is how it is deployed: a clip arrives and the question is what is happening, with "no animal" a legitimate answer. Inputs are frozen per-frame features plus two changed-pixel motion channels; the head is a 0.6 M-parameter MLP over pooled statistics. A previous attempt at this task collapsed onto the majority classes (45% accuracy, per-class F1 $\approx$0). Table II shows the current model and the one change that mattered.

The gain came from the *representation*, not the head. Replacing CLIP with a self-supervised image backbone or a video backbone moved test macro-F1 by +0.047 and +0.059; the worst configuration of either beat the best CLIP configuration, with no overlap in range. Head capacity, by contrast, did

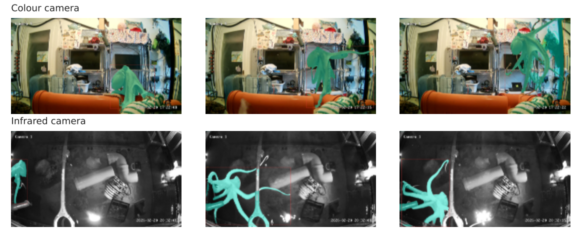

Fig. 2. Mask student output. Per-frame prompting alone bleeds into bright metal tools under infrared and into background clutter on colour cameras; confidence-seeded temporal propagation with a largest-blob constraint removes both, and correctly rejects the reflection camera.

nothing: sweeping hidden width 256–1024 moved validation by +0.0017 against a seed standard deviation of 0.0040, and the widest head was the worst on both backbones.

A second representation effect isolates *animal pixel resolution* as the operative variable. The median animal occupies a bounding box 0.216 of the frame across, so a backbone at 224^2 sees it at $\approx 48 \times 48$ pixels. Three preprocessing variants separate the causes: the default centre crop retains 43% of the frame but preserves animal scale (0.578 validation); letterboxing retains all of it but shrinks the animal (0.574); cropping to the segmentation mask magnifies the animal $\approx 3\times$ and reaches 0.618. Field of view and animal scale trade off almost exactly—only magnification helps.

Segmentation. A 3.2 M-parameter LR-ASPP student distilled from Grounding DINO+SAM2 reaches IoU 0.642 mean / 0.719 median against 122 human-drawn masks on five videos held out of all training, with mean area error $\approx$1% (Fig. 2). It does not meet the 0.85 IoU we set out to reach; we report it because the shortfall is informative. Everything downstream—presence, body area as a posture proxy, motion inside the mask—is area-based, and area is accurate to 1% while boundary quality is what IoU penalises. Two levers produced the gain, both mundane and both measured leak-free: raising input resolution from 256^2 to 512^2 (at 256^2 the median animal is $\approx 40 \times 40$ pixels and an arm is 1–2 pixels wide, i.e. unrepresentable) and replacing symmetric Dice+BCE with Focal–Tversky, which penalises the under-segmentation the symmetric loss ignores.

The mask as a presence gate, counted by video. Mask area is also the better presence test, and measuring it exposed a defect in our own benchmark. Negative sets are never pooled, because the two failure modes are different: an empty tank, and the reflection camera in which the glass shows the room and a mirrored human. Against 105 empty frames drawn from 53 recordings the student separates present from absent at AUC 0.917 (CI_{95} $[0.839, 0.962]$, false positives 0.143 at the deployed area gate); against 34 reflection frames from 27 recordings, 0.921 ($[0.826, 0.966]$, 0.176). The intervals almost entirely overlap, so *neither failure mode dominates*—a null result, and one we had earlier got wrong in both directions. An initial estimate of 0.794 came from 19 negative frames of which 18 were the same recording; it was not merely under-powered but pessimistic, dominated by one unusually hard video. Counting negatives by *video* rather than by frame is the same discipline already applied to the training splits, and applying it late cost us a published number that understated the model.

Temporal fusion does not pay. Smoothing the probability map across neighbouring frames is the obvious way to exploit video, and it fails instructively. Exponential moving average costs mask quality (IoU $0.642 \rightarrow 0.547$) and optical-flow warping is worse on *both* axes (0.511), so what little it buys comes from plain smoothing rather than motion compensation. Its apparent presence gain was large only against the one-recording baseline above; measured against the properly powered 0.917 the remaining headroom is under 0.08. Test-time fusion therefore trades mask quality for almost nothing. A temporally *trained* student—rather than a smoothed per-frame one—remains untested and is the honest next step, since per-frame mislocalisation is the failure mode.

Captioning. The teacher's captions were initially poor on this footage, and the fix was its *input* rather than its size: enhancing frame contrast, raising resolution, and sending only the frames the presence probe scores highest rather than a uniform sample. In a blind A/B over 33 clips the enhanced-input captions were preferred 23 times against 1 (70%, CI_{95} $[53\%, 83\%]$; the remainder ties or both judged poor)—the same lesson as the letterbox and resolution results above, arriving a third time. A 2B VLM adapted with LoRA on those captions then improves held-out embedding similarity from 0.702 to 0.834 and ROUGE-L from 0.269 to 0.455 over its own base model. Quantised to four bits it is $1.7\,$GB and captions a clip in $\approx 3\,$s on a $16\,$GB laptop with no GPU. The released corpus carries ≈25,000 teacher captions, five per clip from disjoint frame samplings, against the 3,066 examples this student was trained on.

Cost and footprint. Teacher cost is linear in frames sent ($\mathrm{tokens} \approx 290 + 386n$), measured at \$0.000603 per six-frame call. The five-pass labelling of the 5,222 surviving clips is therefore 26,110 calls, about \$16. The same five-pass protocol applied to all ≈160,000 clips in the corpus would be 800,000 calls, about \$480, and—at the 13–36 clips/min the hosted model sustains—the 372–1,030 h of wall-clock from Sec. III. The selection cascade is what turns that into a job that finishes: a $\approx 31\times$ reduction in teacher calls, paid for by gates that are essentially free. What is deployed afterwards contains no frontier model at all—a $0.6\,$M-parameter ethogram head, a $3.2\,$M-parameter mask student, and a 2B captioner quantised to $1.7\,$GB that labels a clip in $\approx 3\,$s on a $16\,$GB laptop with no GPU and no network. The teacher is a one-off offline expense; the recurring cost is a laptop.

What did not work. Recorded because each is a natural next thing to try. Feature-space augmentation (mixup, Gaussian noise) is *negative* and monotonically worse with strength; the model is not overfitting-limited, so a regulariser only corrupts frozen features. Class rebalancing hurt the class it was meant to protect—removing it improved the weakest class from F1 0.37 to 0.44. And our own hypothesis that teacher-label quality was the segmentation ceiling was wrong: the evidence for it turned out to be train leakage, and on a clean holdout, mask IoU is flat across teacher quality (0.494, 0.508, 0.506).

TABLE III
DISTILLED STUDENTS VS. ZERO-SHOT TEACHERS

task	system	metric
mask	GD+SAM2, zero-shot, per frame	IoU 0.374
	ours, $3.2\,$M, no prompt	IoU $\mathbf{0.642}$
	paired $\Delta = -0.268$, CI_{95} $[-0.313, -0.136]$	
presence	zero-shot CLIP, 5+5 prompts	AUC 0.450
	ours, trained probe	AUC $\mathbf{0.745}$
	paired $\Delta = +0.295$, CI_{95} $[+0.069, +0.528]$	
caption	base 2B VLM	emb-sim 0.702
	ours, LoRA	emb-sim $\mathbf{0.834}$

TABLE IV
COMPARISON WITH HIDEANDSEG [14]

	[14]	our teacher	**our student**
mask IoU	0.938	0.726^{a}	$\mathbf{0.642}$
reference	init. frame	human masks	human masks
prompt at inference	auto	gated	**none**
presence	given	inferred	**inferred**
params	≈249 M	≈230 M	$\mathbf{3.2}\,$M
data/code released	no	–	**yes**
setting	wild, 3 sites	1 tank	1 tank
scale	148 vid.	–	4,665 clips

Table III sets the students against the zero-shot models they were distilled from, on identical held-out human labels; paired differences are bootstrapped clustered by source video and both intervals exclude zero. The two presence arms share the frozen backbone *and* its preprocessing, so the gap isolates the trained probe rather than the features. One caveat travels with the mask row: on the 21 of 122 frames where Grounding DINO clears its own confidence gate the teacher *wins*, 0.726 against 0.657. The teacher is high-precision and low-recall—it finds nothing at all on a quarter of frames—and distillation converted a sparse, high-quality signal into dense coverage. That is a better argument for distilling than a flat win would have been.

Table IV places us beside that work. Reported mask quality tracks how much prompting is assumed at inference, and every other row differs, so the IoU row is not a ranking. We claim the unprompted, presence-inferring and $\approx 78\times$-smaller rows, and the release; we do not claim the IoU row, since neither method has been run on the other's data and the final two rows favour them.

V. WHAT THE PIPELINE REVEALS

Running the pipeline over the corpus yields a behavioural record, which is the reason to build it (Fig. 3). Across 3,083 clips with a visible animal, the activity budget is 41% ex-

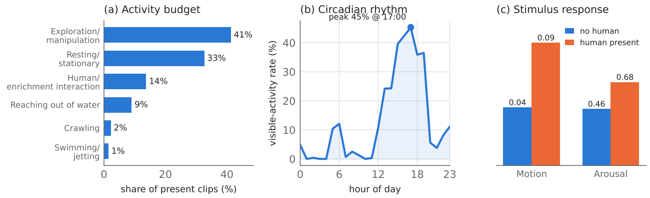

Fig. 3. Behaviour recovered from the corpus: (a) activity budget, with locomotion split into its crawling and swimming/jetting components; (b) exposure-normalised visible activity by hour of day—normalising by the number of windows examined per hour is essential, since without it the profile tracks recording coverage rather than the animal; and (c) mean motion and arousal with and without a human present.

ploration/manipulation, 33% resting, 14% human/enrichment interaction, 9% reaching out of water and 3% locomotion.

Circadian structure. Normalising visible activity by the number of windows actually examined in each hour—without which the profile simply tracks recording coverage—gives a strongly diurnal animal: 1–5% visible activity overnight rising to a 45% peak at 17:00, with a plateau from 13:00 to 19:00 and a smaller rise at dawn.

Response to human presence. Clips the extractor labels as containing a human show nearly double the absolute motion of baseline clips (0.095 against 0.045) and a higher score on a transparent arousal rubric (0.68 against 0.46). On the colour cameras, the most common body colour at baseline (dark red-brown, 16% of clips) is markedly less frequent during human interaction (6%).

Posture and location. The structured record carries more than a class per clip. Of the 3,083 clips with a visible animal, 64% show arms extended against 18% contracted, and 53% place the animal on open substrate, 20% on the glass and 19% at the surface. Only 7.5% place it in the den or at its entrance—which is a statement about what the pipeline can see, not about the animal. Time spent hidden in the den is exactly what the presence gate discards, so den occupancy is under-observed by construction and the denominator here is visible time, not elapsed time.

Viewpoint is a confound, not a finding. Behaviour composition differs sharply by camera: the rear view is dominated by human interaction and reaching out of water, the front view by exploration. This is where the cameras point, not how the animal differs, and it is a reason to pool cameras before comparing behaviours rather than to report per-camera profiles as results.

Two caveats bound these. The tank contains permanent enrichment objects, so a context label of "enrichment object present" fires on ≈66% of clips and means only that—not active enrichment; the clean contrast is human present against absent. And the upstream presence gate is imperfect, so absolute levels will shift with a better detector while the *rate contrasts* above, which are ratios within the same gate, are robust to it.

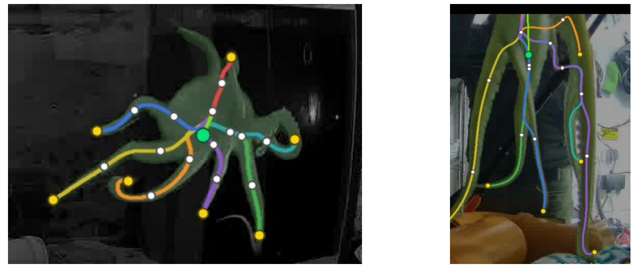

Fig. 4. Skeletonisation on two clips, each recovering seven of eight arms with a correct head. Colour distinguishes individual arms; every node is marked detected, fitted or occluded so that kinematics are computed only from evidence-backed samples. Left: an infrared frame during human interaction. Right: the animal against the tank glass.

VI. Released Data and Benchmarks

No public annotated octopus behaviour corpus exists. We release, at github.com/sidraj000/octopus-behaviour: 5,222 clips with five-pass teacher labels *including the full vote distribution* rather than only the majority; ≈25,000 captions (5,160 clips × five disjoint frame samplings), which additionally permits measuring how a VLM's description of one behaviour varies with which frames it happens to see; a frozen six-class ethogram dataset of 4,665 clips over 204 videos with video-level splits and per-clip reliability weights; 456 human behaviour labels; 513 human masks; and 3,205 structured behavioural records of nine fields each. Three benchmarks are frozen: 122 human masks on five held-out videos; 50 human arm-keypoint frames over 20 videos; and negative sets counted by recording rather than by frame—105 empty-tank frames from 53 recordings and 34 reviewed reflection frames from 27.

Four protocol rules are part of the release, because the pipeline broke each of them at least once during development: splits are by source video, never by clip; frozen sets are never regenerated to suit a result; negatives of different kinds are never pooled; and holdout videos are excluded from *every* training source, not merely the final stage.

VII. Skeletonisation: Toward Limb-Level Ethograms

The mask student also supports a finer representation than a class label. We skeletonise the silhouette into an anatomical graph—mantle, head and eight arms—by thinning and shortest-path tracing, tracking it across frames with an optical-flow prior and marking every node *detected*, *fitted* or *occluded* so that kinematics use only evidence-backed samples (Fig. 4). On 50 human-annotated frames the arm-tip F1 is 0.539 (precision 0.72) at 3.68 arms recovered per frame; seeding propagation from the best-resolved frame rather than the first was worth +3.67 arms. Speeds are in pixels per second in crop space, uncalibrated, and arm counts are silhouette-limited.

Two independent halves of the pipeline agree. The skeleton is worth more than a capability claim, because it can be checked against the ethogram without either signal seeing

TABLE V
ARM-TIP SPEED BY BEHAVIOUR CLASS, AGGREGATED BY VIDEO

behaviour	median (px/s)	videos
resting / stationary	53	24
human/enrichment interaction	90	24
crawling	91	19
swimming / jetting	107	15
exploration / manipulation	112	25
reaching out of water	141	25

the other: the kinematics pipeline never reads the behaviour label, and the teacher never sees a mask. On a video-spread sample of 146 clips from 66 recordings, aggregated to one value per (video, class) before testing so that several clips from one recording cannot masquerade as independent evidence, raw arm-tip speed orders the classes as an ethologist would (Table V). The separation is not marginal—Kruskal–Wallis $H = 33.2$ on $N = 132$ video-class units, $p = 3.5 \times 10^{-6}$, $\varepsilon^2 = 0.224$—and all five resting-versus-other contrasts survive Holm correction ($p \leq 0.0086$). A purely geometric measurement derived from the mask thus recovers the behavioural distinction a language model was asked about independently, which is the strongest evidence we have that the pipeline measures the animal rather than its own conventions.

Its further purpose is the obvious next question: a six-class sheet cannot distinguish exploration with one arm from whole-body exploration, and per-limb usage statistics would make *limb-aware* ethograms possible. That is the direction we intend to take.

VIII. LIMITATIONS

One animal, one tank. Every measurement is from a single adult *O. vulgaris*. Footage for four further animals, including 155 days of a second individual, is available and unprocessed; cross-animal generalisation is untested, and this is the paper's principal limitation.

Agreement, not accuracy. All 456 human behaviour labels were collected with the model's suggestion visible, so they measure agreement with the model, not independent accuracy. On the subset where the two disagree, the human sides with the teacher over the student roughly 3:2. A blind round on the reserved test videos is specified but not yet run, and until it is, the ethogram figures should be read as teacher reproduction.

Selection on a small validation set. Roughly six design choices were made on 35 validation videos, and validation and test disagreed on three of them. Reported margins are softer than their seed variance suggests; we report the validation-selected configuration throughout rather than the best test score.

Infrared. The colour-trained mask student is unusable on the infrared camera, which is 35% of the corpus, so mask-derived quantities are available for colour cameras only.

IX. CONCLUSION

Continuous behavioural monitoring of a captive cephalopod is achievable if the expensive model is used to teach rather than to run. Placing each gate in the cascade by measurement rather than intuition mattered more than any single model choice: every gate ended up somewhere other than the obvious default, and the largest single improvement to behaviour classification came from changing the frozen representation and how much of the animal it sees, not from any change to the classifier. The resulting local models recover circadian structure and a clear response to human presence from footage that no one scored by hand. Data, benchmarks and code are released; the immediate next steps are a blind human round on the reserved videos and the second animal.

ETHICAL STATEMENT AND DATA AVAILABILITY

All footage is observational, from cameras already installed for husbandry, with no intervention or manipulation of the animal. The labels, frozen benchmarks, code, review interfaces and the full experimental record—including the experiments that failed and the conclusions we retracted—are at github.com/sidraj000/octopus-behaviour, under Apache-2.0 for code and CC-BY-4.0 for the labels. The trained models and the larger datasets are linked from there. The footage itself is not ours to redistribute, but every label carries the source video and time offsets needed to reconstruct its clip.

REFERENCES

[1] K. Patel, A. Sharma, H. Kumar, P. Kurundodi, H. Burgsteiner, and W. Slany, "A Novel Framework for Decoding Inter-Species Communication with Octopodes (DISC-O)," in *OCEANS 2025*, pp. 1–9, Sep. 2025, doi: 10.23919/OCEANS59106.2025.11244956.

[2] A. Radford *et al.*, "Learning Transferable Visual Models from Natural Language Supervision," in *ICML*, 2021.

[3] S. Liu *et al.*, "Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection," *arXiv:2303.05499*, 2023.

[4] N. Ravi *et al.*, "SAM 2: Segment Anything in Images and Videos," *arXiv:2408.00714*, 2024.

[5] E. J. Hu *et al.*, "LoRA: Low-Rank Adaptation of Large Language Models," in *ICLR*, 2022.

[6] A. Howard *et al.*, "Searching for MobileNetV3," in *ICCV*, 2019.

[7] Qwen Team, "Qwen3-VL," 2025.

[8] M. Minderer, A. Gritsenko, and N. Houlsby, "Scaling Open-Vocabulary Object Detection," in *NeurIPS*, 2023.

[9] A. Mathis, P. Mamidanna, K. M. Cury, T. Abe, V. N. Murthy, M. W. Mathis, and M. Bethge, "DeepLabCut: markerless pose estimation of user-defined body parts with deep learning," *Nature Neuroscience*, vol. 21, no. 9, pp. 1281–1289, 2018.

[10] T. D. Pereira *et al.*, "SLEAP: A deep learning system for multi-animal pose tracking," *Nature Methods*, vol. 19, no. 4, pp. 486–495, 2022.

[11] L. Borrelli, F. Gherardi, and G. Fiorito, *A Catalogue of Body Patterning in Cephalopoda*. Florence, Italy: Firenze University Press, 2006.

[12] J. A. Mather and J. S. Alupay, "An ethogram for benthic octopods (Cephalopoda: Octopodidae)," *Journal of Comparative Psychology*, vol. 130, no. 2, pp. 109–127, 2016.

[13] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," arXiv:1503.02531, 2015.

[14] A. de Aguiar, M. P. Andrade, C. M. D. Santos, and J. P. Gois, "HideAndSeg: an AI-based tool with automated prompting for octopus segmentation in natural habitats," *arXiv:2511.04426*, Nov. 2025.